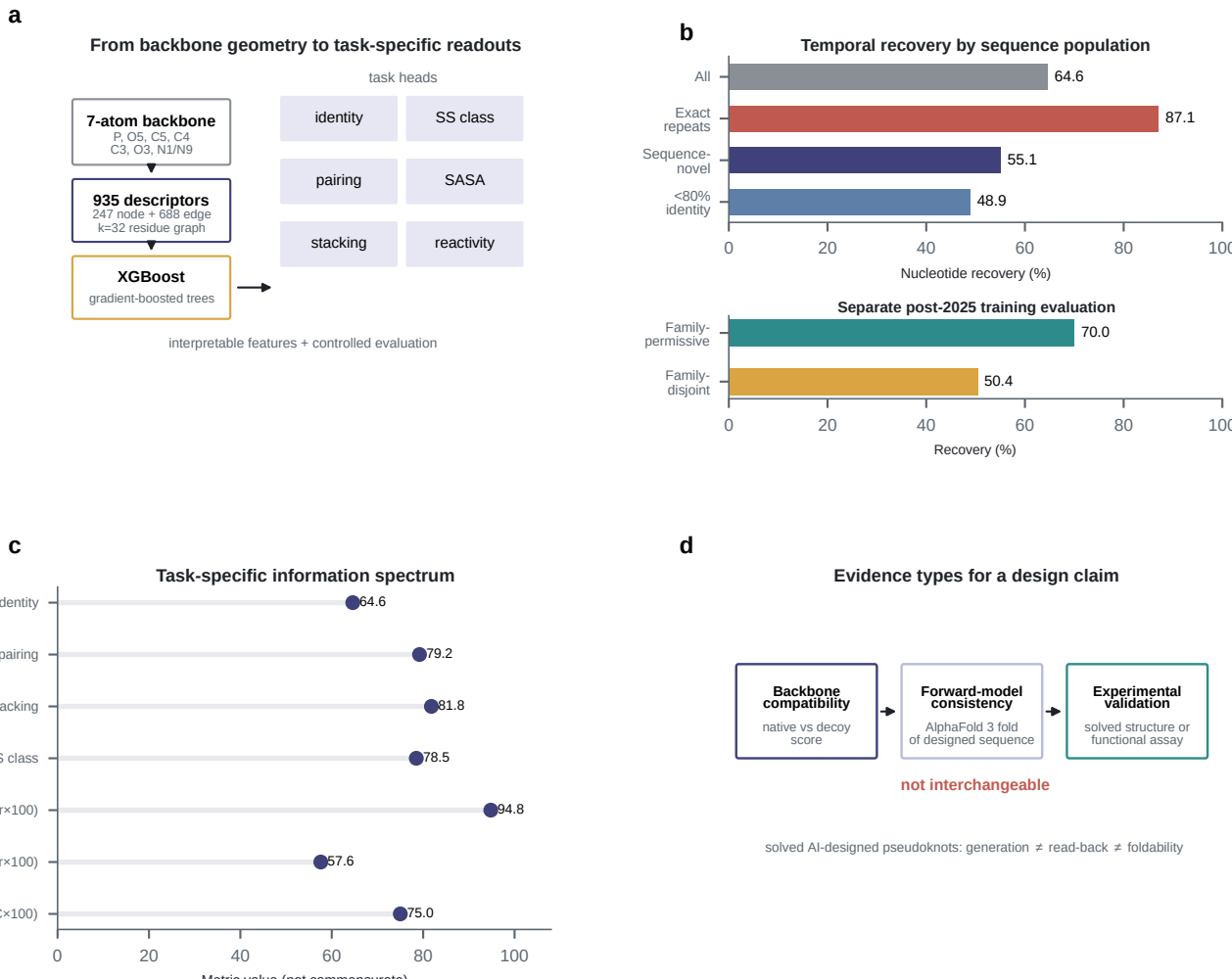
a
From backbone geometry to task-specific readouts
task heads
7-atom backbone
P, O5, C5, C4
C3, O3, N1/N9
935 descriptors
247 node + 688 edge
k=32 residue graph
XGBoost
gradient-boosted trees
identity
SS class
pairing
SASA
stacking
reactivity
interpretable features + controlled evaluation
b
Temporal recovery by sequence population
All
64.6
Exact repeats
87.1
Sequence-novel
55.1
<80% identity
48.9
0
20
40
60
80
100
Nucleotide recovery (%)
Separate post-2025 training evaluation
Family-permissive
70.0
Family-disjoint
50.4
Recovery (%)
c
Task-specific information spectrum
identity
64.6
pairing
79.2
stacking
81.8
SS class
78.5
SASA (r×100)
94.8
reactivity (r×100)
57.6
pseudoknot (AUC×100)
75.0
Metric value (not commensurate)
d
Evidence types for a design claim
Backbone compatibility
native vs decoy score
Forward-model consistency
AlphaFold 3 fold of designed sequence
Experimental validation
solved structure or functional assay
not interchangeable
solved AI-designed pseudoknots: generation ≠ read-back ≠ foldability